



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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24th September 2026

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- Methodology
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Executive Summary

- Summary of methodologies
 - Data Collection through API
 - Data Collection with Web Scraping
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 - Interactive Visual Analytics with Folium
 - Machine Learning Prediction
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 - Predictive Analytics results

Introduction

- **Project background and Context**

- SpaceX advertises Falcon 9 rocket launches on its website with a cost of 62 million dollars; other providers cost upward of 165 million dollars each, much of the saving is because Space X can reuse the first stage. Therefore if we can determine if the first stage will land, we can determine the cost of a launch. This information can be used if an alternate company wants to bid against space X for a rocket launch. This goal of the project is to create a machine learning pipeline to predict if the first stage will land successfully.

- **Problems you want to find answers**

- What factors determine if the rocket will land Successfully?
- The interaction amongst various features that determine the success rate of a successful landing.
- What operation conditions needs to be in place to ensure a successful landing program.

Section 1

Methodology

Methodology

Executive Summary

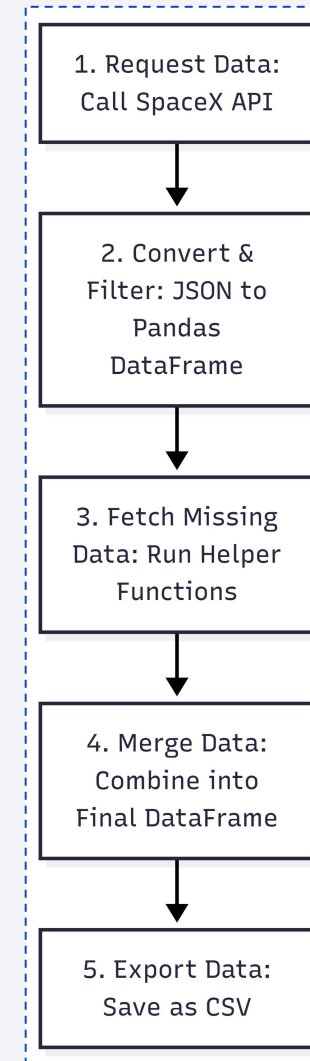
- **Data collection methodology:**
 - Data was collected using SpaceX API and Web Scraping from Wikipedia
- **Perform data wrangling**
 - One-hot encoding was applied to categorical Features
- **Perform exploratory data analysis (EDA) using visualization and SQL**
- **Perform interactive visual analytics using Folium and Plotly Dash**
- **Perform predictive analysis using classification models**
 - How to build, tune, evaluate classification models

Data Collection

- **The Data was collected using various methods**
 - Data collection was done using get request to the SpaceX API.
 - Next, we decoded the response content as JSON using `.json()` function call and turn it into a pandas dataframe using `.json_normalize()`.
 - We then cleaned the data, checked for missing values and fill in missing values where necessary
 - In Addition, we performed web scarping from wikipedia for Falcon 9 launch records with BeautifulSoup.
 - The Objective was to extract the launch records as HTML table, parse the table and convert it to a pandas dataframe for future analysis.

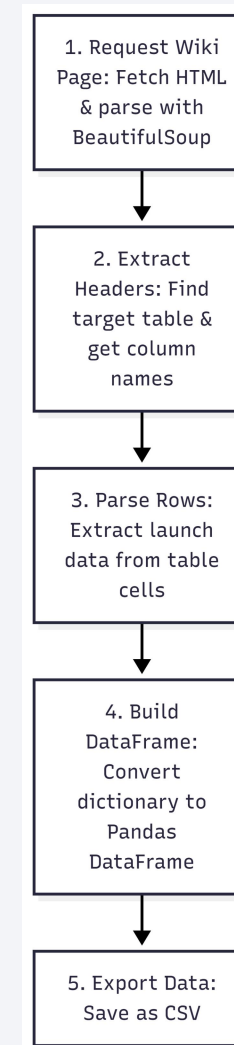
Data Collection – SpaceX API

- We used the get request to the SpaceX API to collect data, clean the requested data and did some basic Data wrangling and formatting.
- The GitHub URL of the completed SpaceX API calls notebook (https://github.com/Anshulworld/IBM-Data-Science-Capstone-SpaceX-Project/blob/main/LAB1_Data_Collection_API_SpaceX.ipynb), as an external reference and peer-review purpose
- LAB1_Data_Collection_API_SpaceX.ipynb

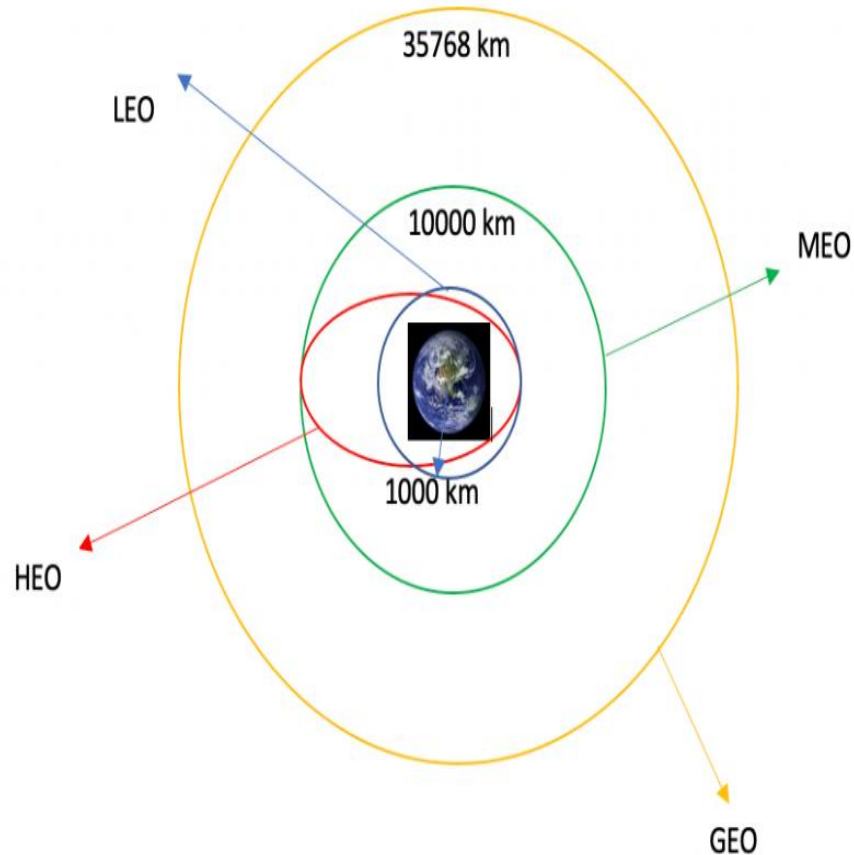


Data Collection - Scraping

- We Applied web scraping to webscrap Falcon 9 launch records with BeautifulSoup
- We parsed the table and converted it into a pandas dataframe.
- The GitHub URL of the completed web scraping notebook is:
https://github.com/Anshulworld/IBM-Data-Science-Capstone-SpaceX-Project/blob/cd91beaa11ec9d20ba354a82b6cdbf3e42a27ad3/LAB2_Data_Collection_with_Web_Scraping_SpaceX.ipynb
- LAB2_Data_Collection_with_Web_Scraping_SpaceX.ipynb



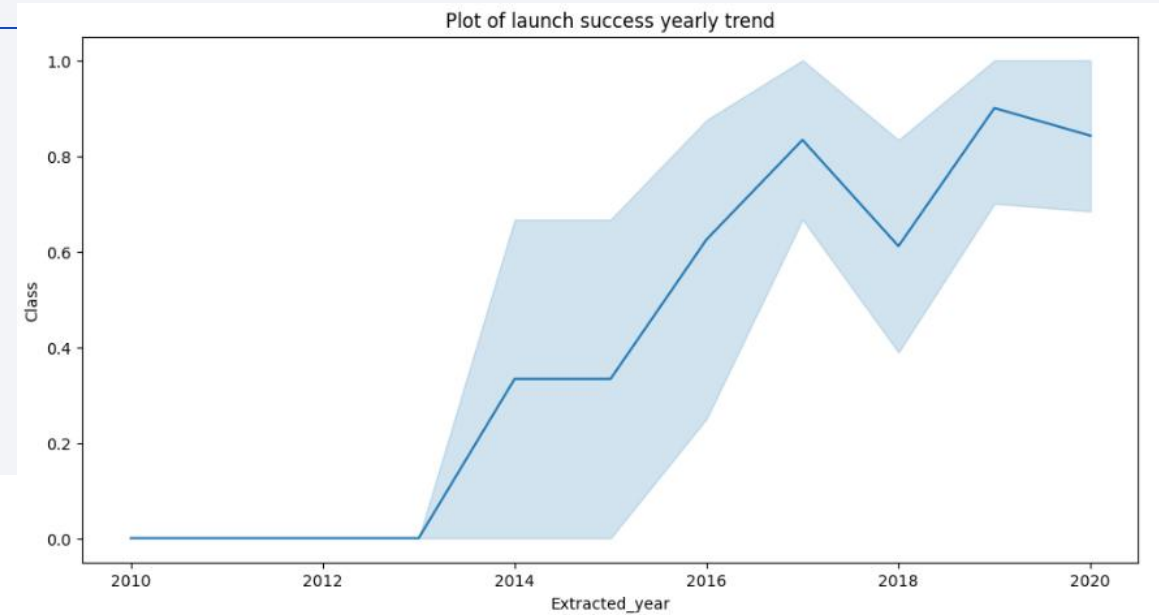
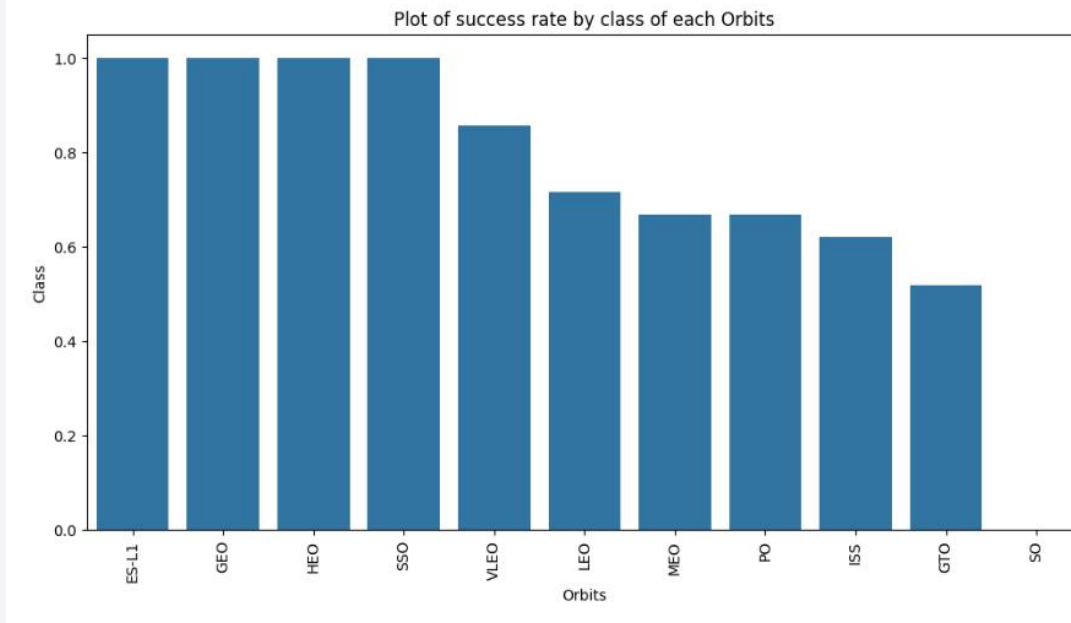
Data Wrangling



- We performed exploratory data analysis and determined the training labels.
- We calculated the number of launches at each site, and the number and occurrence of each orbit.
- We created landing outcomes label from outcome column and exported the results to CSV.
- The GitHub URL of my completed data wrangling related notebooks is https://github.com/Anshulworld/IBM-Data-Science-Capstone-SpaceX-Project/blob/cd91beaa11ec9d20ba354a82b6cdbf3e42a27ad3/LAB3_Data_Wrangling_SpaceX.ipynb

EDA with Data Visualization

- We Explored the data by visualizing the relationship between flight number and launch site, payload and launch site, success rate of each orbit type, flight number and orbit type, the launch success yearly trends



The link to the notebook is

https://github.com/Anshulworld/IBM-Data-Science-Capstone-SpaceX-Project/blob/main/LAB5_EDA_with_Data_Visualization.ipynb

EDA with SQL

- We loaded the SpaceX dataset into SQLite Database without leaving the jupyter notebook.
- We applied EDA with SQL to get insight from the data. We wrote queries to find out instance:
 - *the names of unique launch sites in the space mission*
 - *The total payload mass carried by booster launched by NASA (CRS)*
 - *The average payload mass carried by booster version f9 v1.1*
 - *The total number of successful and failure mission outcomes*
 - *The failed landing outcomes in drone ship, their booster version and launch site names.*
- The link to the notebook is https://github.com/Anshulworld/IBM-Data-Science-Capstone-SpaceX-Project/blob/main/LAB4_EDA_with_SQL.ipynb

Build an Interactive Map with Folium

- We marked all launch sites, and added map objects such as markers, circles, lines to mark the, success or failure of launches for each site on the folium map.
- We assigned the feature launch outcomes (failure or success) to class 0 and 1 .i.e. , 0 for failure, and 1 for success.
- Using the color-labeled marker clusters, we indentified which launch sites have relatively high success rate.
- We calculated the distances between a launch site to its proximities.
We answered some question for instance:
 - Are launch sites near railways, highways and coastlines.
 - Do launch sites keep certain distance away from cities.

Build a Dashboard with Plotly Dash

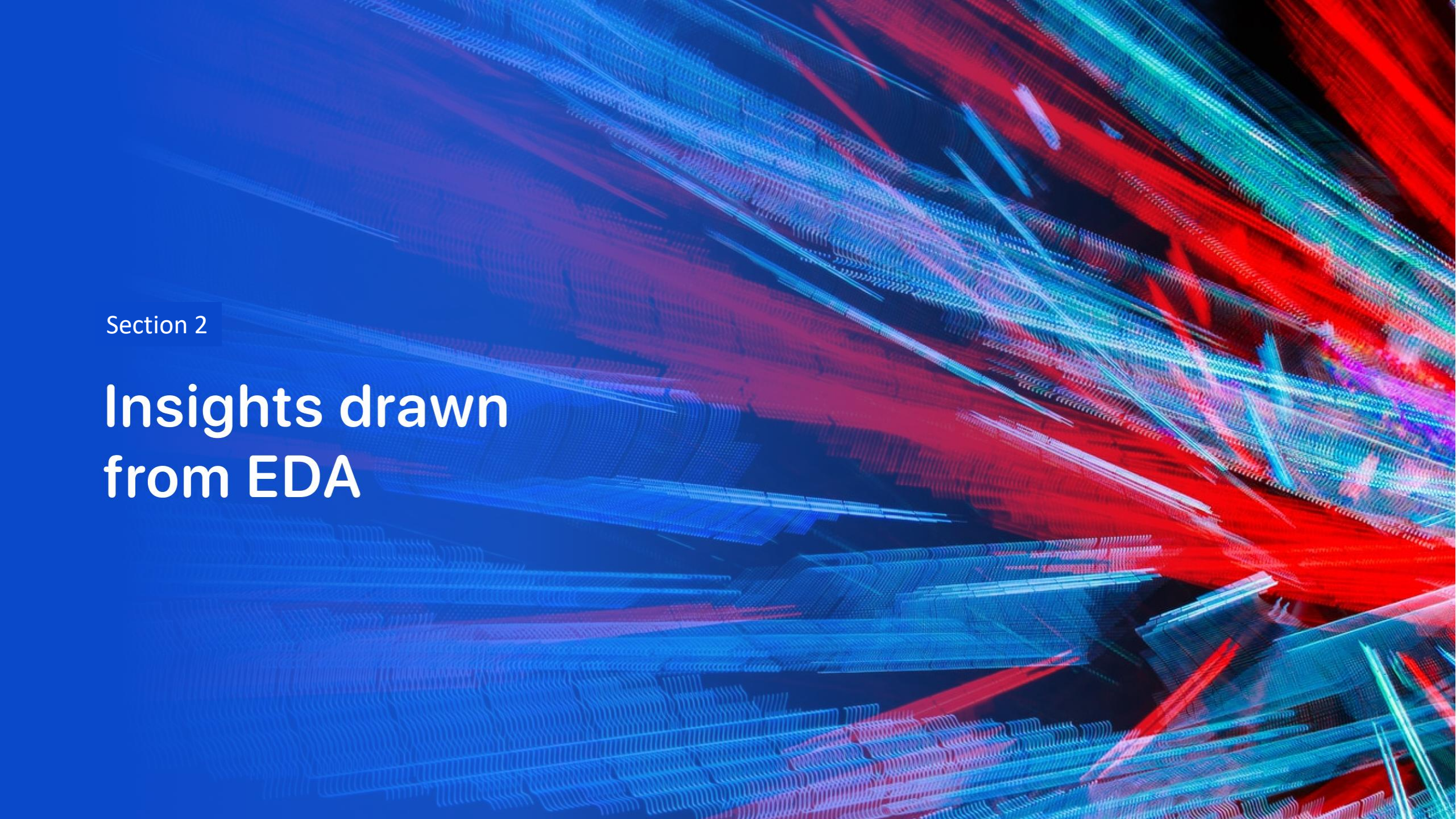
- We built an interactive dashboard with Plotly Dash
- We plotted pie charts showing the total launches by a certain site
- We plotted a scatter graph showing the relationship with outcome and payload Mass (kg) for the different booster version.
- The link to the notebook is https://github.com/Anshulworld/IBM-Data-Science-Capstone-SpaceX-Project/blob/main/LAB7_SpaceX_Dashboard/spacex-dash-app.py

Predictive Analysis (Classification)

- We loaded the data using numpy and pandas, transformed the data, split out data into training and testing.
- We built different machine learning models and tune different hyperparameters using GridSearchCV
- We used accuracy as the metric for our model, improved the model using feature engineering and algorithm tuning.
- we found the best performing classification model.
- The link to the notebook is https://github.com/Anshulworld/IBM-Data-Science-Capstone-SpaceX-Project/blob/main/LAB8_Machine_Learning_Prediction_Lab_SpaceX.ipynb

Results

- **Exploratory data analysis results**
- **Interactive analytics demo in screenshots**
- **Predictive analysis results**

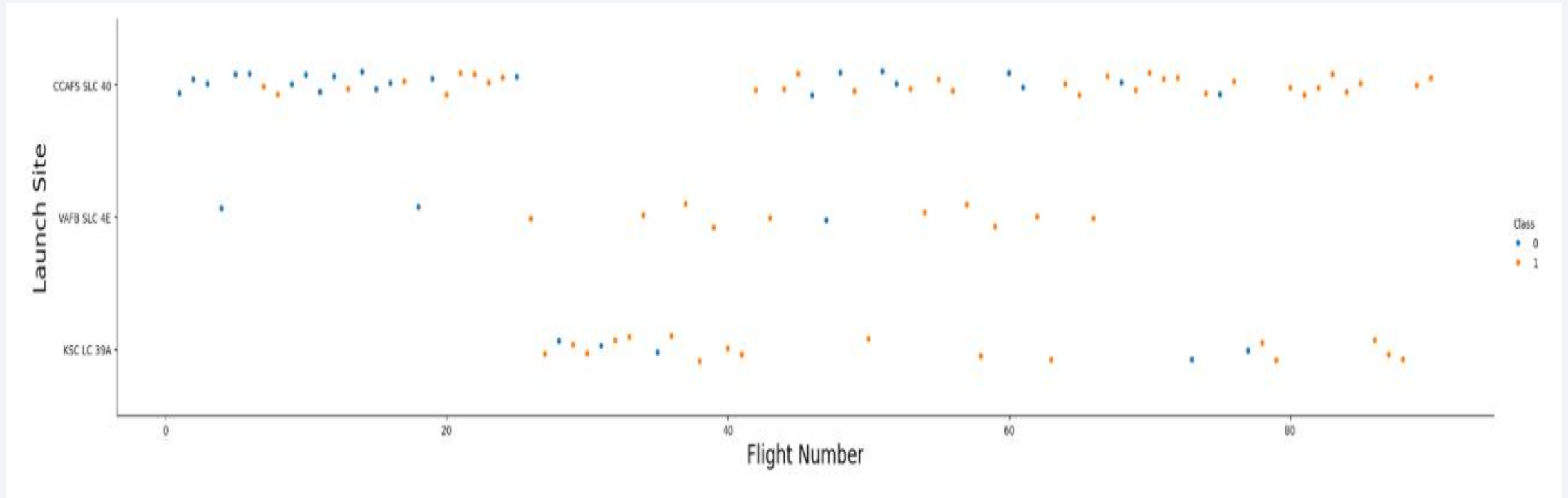
The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is dynamic and technological.

Section 2

Insights drawn from EDA

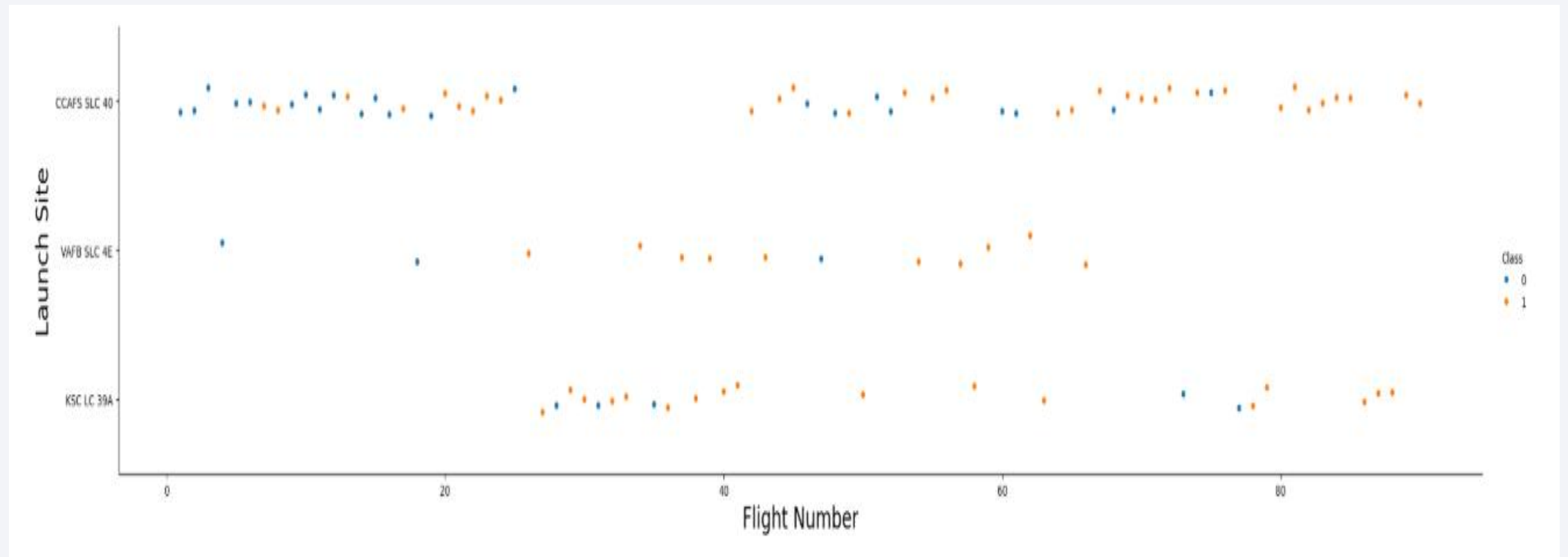
Flight Number vs. Launch Site

- From the plot, we found that the larger the flight amount at a launch site, the greater the success rate at a launch site.



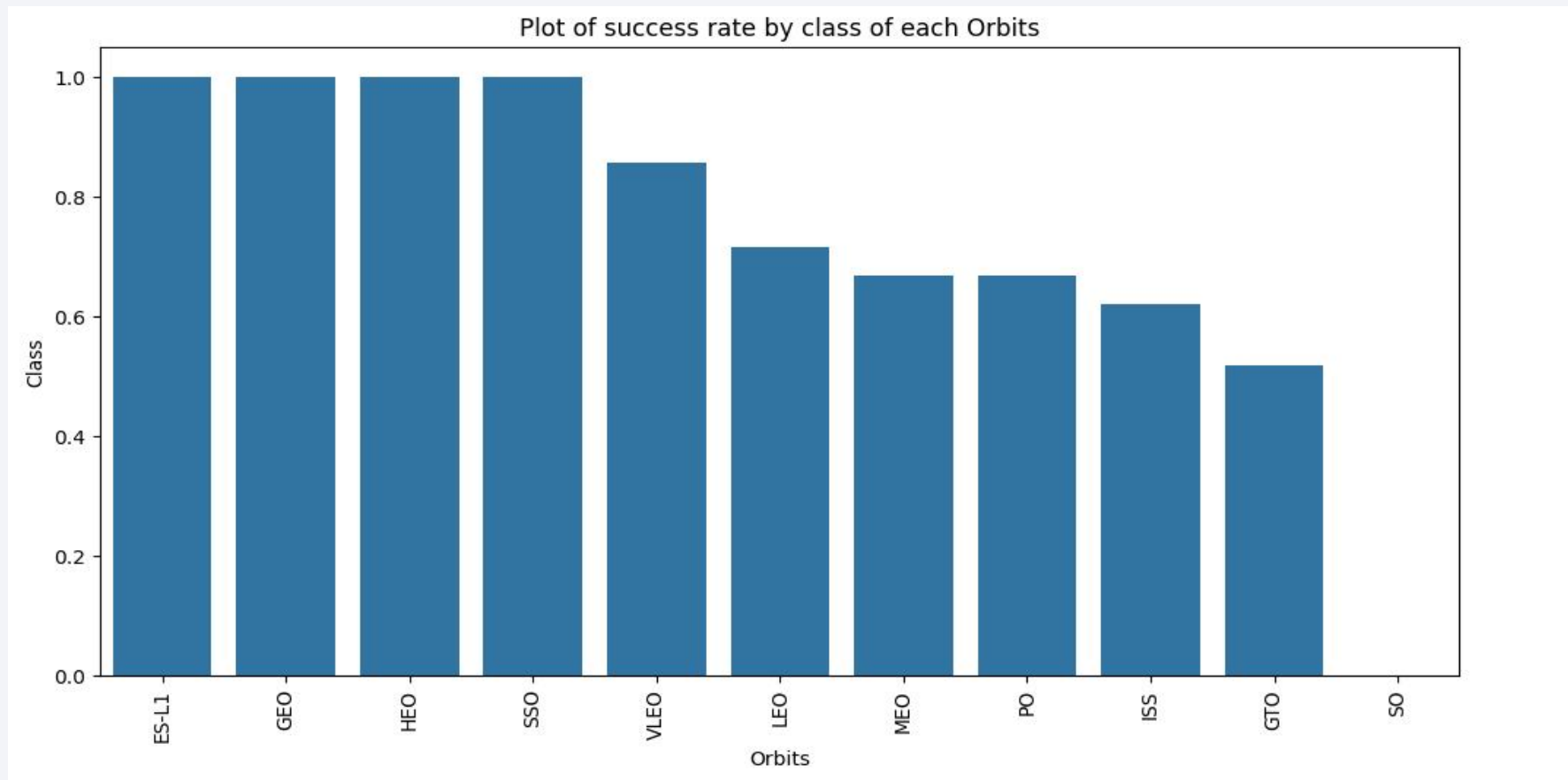
Payload vs. Launch Site

- The Greater the payload mass for launch site CCAFS SLC 40 the higher the success rate for the rocket.



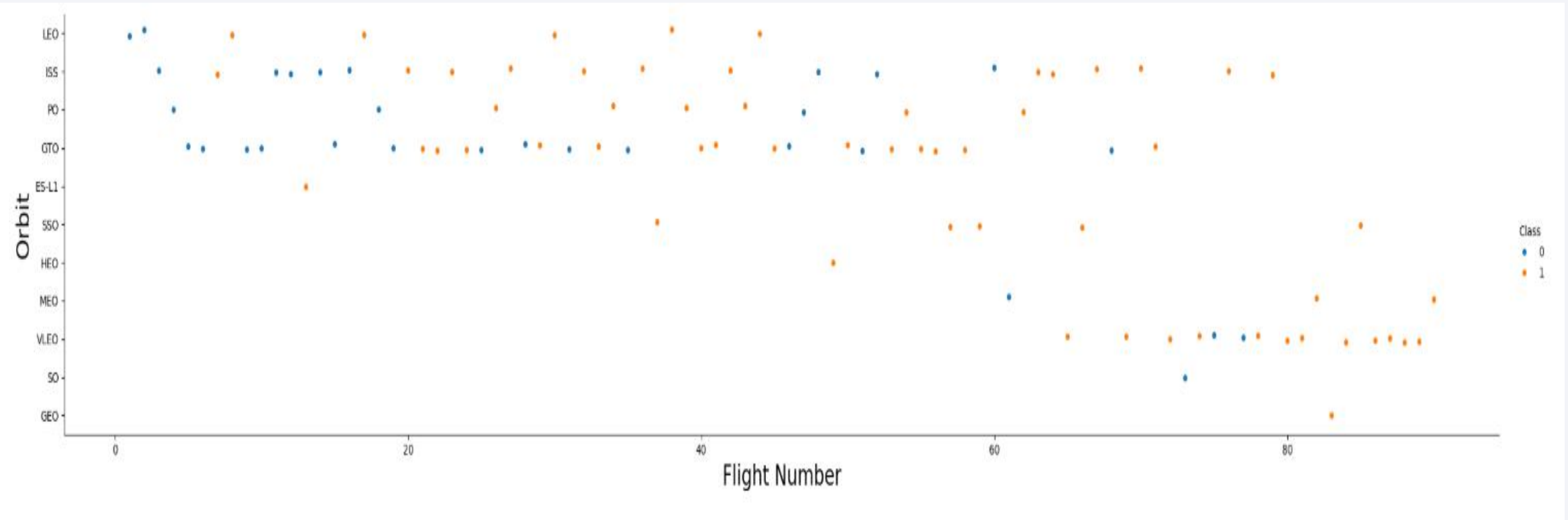
Success Rate vs. Orbit Type

- From the plot, we can see that ES-L1, GEO, HEO, SSO VLEO has the most success rate.



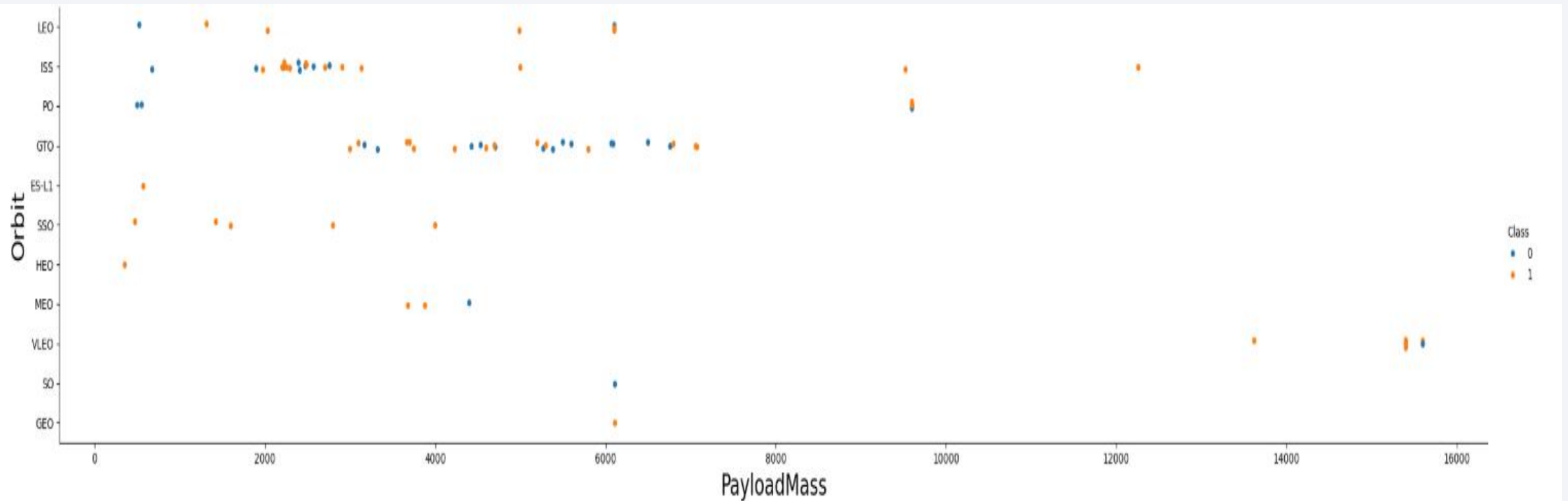
Flight Number vs. Orbit Type

- The plot below shows the Flight Number Vs. Orbit type. We observe that in the LEO orbit, success is related to the number of flights whereas in the GTO orbit, there is no relationship between flight number and the orbit.



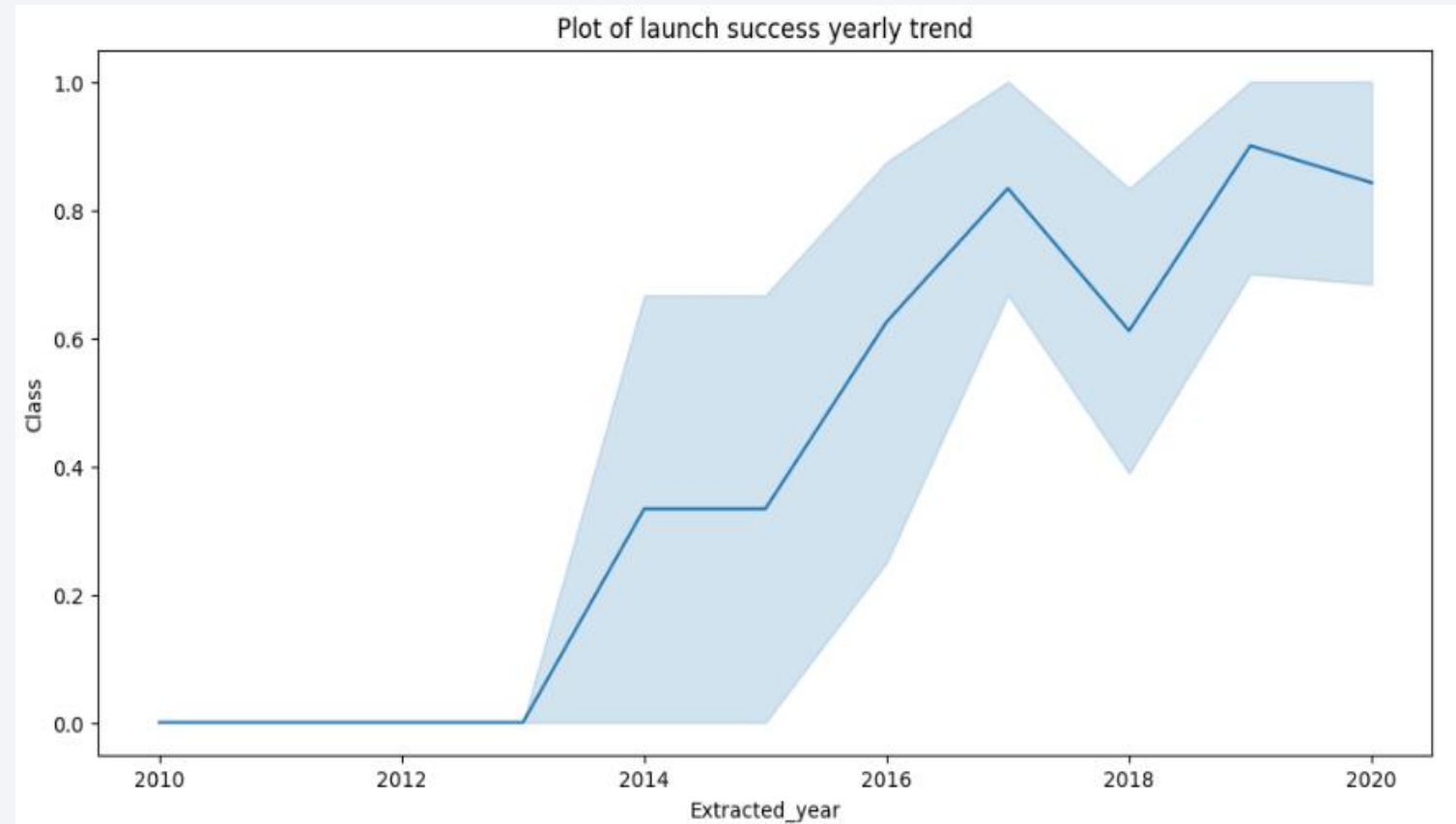
Payload vs. Orbit Type

- We can observe that with heavy payloads, the successful landing are more for PO, LEO and ISS orbits.



Launch Success Yearly Trend

- From the plot, we can observe that success rate since 2013 kept on increasing till 2020.



All Launch Site Names

- We used the key word **DISTINCT** to show only unique launch sites from the SpaceX data.

```
Display the names of the unique launch sites in the space mission

In [12]: %sql SELECT DISTINCT Launch_Site FROM SPACEXTABLE;

* sqlite:///my_data1.db
Done.

Out[12]: Launch_Site
         CCAFS LC-40
         VAFB SLC-4E
         KSC LC-39A
         CCAFS SLC-40
```

Launch Site Names Begin with 'CCA'

Display 5 records where launch sites begin with the string 'CCA'

```
In [13]: %sql SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

* sqlite:///my_data1.db

Done.

```
Out[13]:
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- We Used the query above to display 5 records where launch sites begin with 'CCA'.

Total Payload Mass

- We calculated the total payload carried by boosters from NASA as 48213 using the query below

Display the total payload mass carried by boosters launched by NASA (CRS)

```
In [14]: %sql SELECT SUM(PAYLOAD_MASS_KG_) AS Total_Payload_Mass FROM SPACEXTABLE WHERE Customer LIKE '%NASA (CRS)%';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Out[14]: Total_Payload_Mass
```

```
48213
```

Average Payload Mass by F9 v1.1

- We calculated the average payload mass carried by booster version F9 V1.1 as 2928.4

Display average payload mass carried by booster version F9 v1.1

```
In [15]: %sql SELECT AVG(PAYLOAD_MASS_KG_) AS Average_Payload_Mass FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Out[15]: Average_Payload_Mass
```

```
2928.4
```

First Successful Ground Landing Date

- We observed that the dates of the first successful landing outcomes on ground pad was 22nd December 2015

List the date when the first succesful landing outcome in ground pad was acheived.

Hint: Use min function

```
In [16]: %sql SELECT MIN(Date) AS First_Successful_Landing FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (ground pad)';
```

```
* sqlite:///my_data1.db  
Done.
```

```
Out[16]: First_Successful_Landing  
2015-12-22
```

Successful Drone Ship Landing with Payload between 4000 and 6000

- We used the **WHERE** clause to filter for boosters which have successfully landed on drone ship and applied the **AND** condition to determine successful landing with payload mass greater than 4000 but less than 6000

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
In [17]: %sql SELECT Booster_Version FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000;
```

```
* sqlite:///my_data1.db  
Done.
```

```
Out[17]: Booster_Version
```

```
F9 FT B1022
```

```
F9 FT B1026
```

```
F9 FT B1021.2
```

```
F9 FT B1031.2
```

Total Number of Successful and Failure Mission Outcomes

- We used wildcard like % to filter for **WHERE** Mission Outcome was a success or a failure.

List the total number of successful and failure mission outcomes

```
In [18]: %sql SELECT Mission_Outcome, COUNT(Mission_Outcome) AS Total_Count FROM SPACEXTABLE GROUP BY Mission_Outcome;
```

* sqlite:///my_data1.db

Done.

```
Out[18]:
```

Mission_Outcome	Total_Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- We determined the booster that have carried the maximum payload using a subquery in the **WHERE** clause and the **MAX()** function.

List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

```
In [19]: %sql SELECT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS_KG_ = (SELECT MAX(PAYLOAD_MASS_KG_) FROM SPACEXTABLE);  
* sqlite:///my_data1.db  
Done.
```

```
Out[19]: Booster_Version  
F9 B5 B1048.4  
F9 B5 B1049.4  
F9 B5 B1051.3  
F9 B5 B1056.4  
F9 B5 B1048.5  
F9 B5 B1051.4  
F9 B5 B1049.5  
F9 B5 B1060.2  
F9 B5 B1058.3  
F9 B5 B1051.6  
F9 B5 B1060.3  
F9 B5 B1049.7
```

2015 Launch Records

- We Used a combination of the **WHERE** clause, **LIKE**, **AND**, and **BETWEEN** conditions to filter for failed landing outcomes in drone ship, their booster versions, and launch sites names for year 2015.

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

```
In [20]: %sql SELECT substr(Date, 6, 2) AS Month, "Landing_Outcome", Booster_Version, Launch_Site FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Failure (drone ship)' AND substr(Date, 1, 4) = '2015';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Out[20]:
```

	Month	Landing_Outcome	Booster_Version	Launch_Site
--	-------	-----------------	-----------------	-------------

01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
----	----------------------	---------------	-------------

04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40
----	----------------------	---------------	-------------

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- We selected Landing outcomes and the **COUNT** of landing outcomes from the data and used the **WHERE** clause to filter for landing outcomes **BETWEEN** 2010-06-04 to 2017-03-20.
- We applied the **GROUP BY** clause to group the landing outcomes and the **ORDER BY** clause to order the grouped landing outcome in descending order.

Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

```
In [22]: %sql SELECT "Landing_Outcome", COUNT("Landing_Outcome") AS Outcome_Count FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY "Landing_Outcome" ORDER BY Outcome_Count DESC;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Out[22]:
```

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a deep blue, with the horizon line visible. The city lights are concentrated in the lower right quadrant, showing a dense network of urban areas. The text "Section 3" is overlaid on the left side of the image.

Section 3

Launch Sites Proximities Analysis

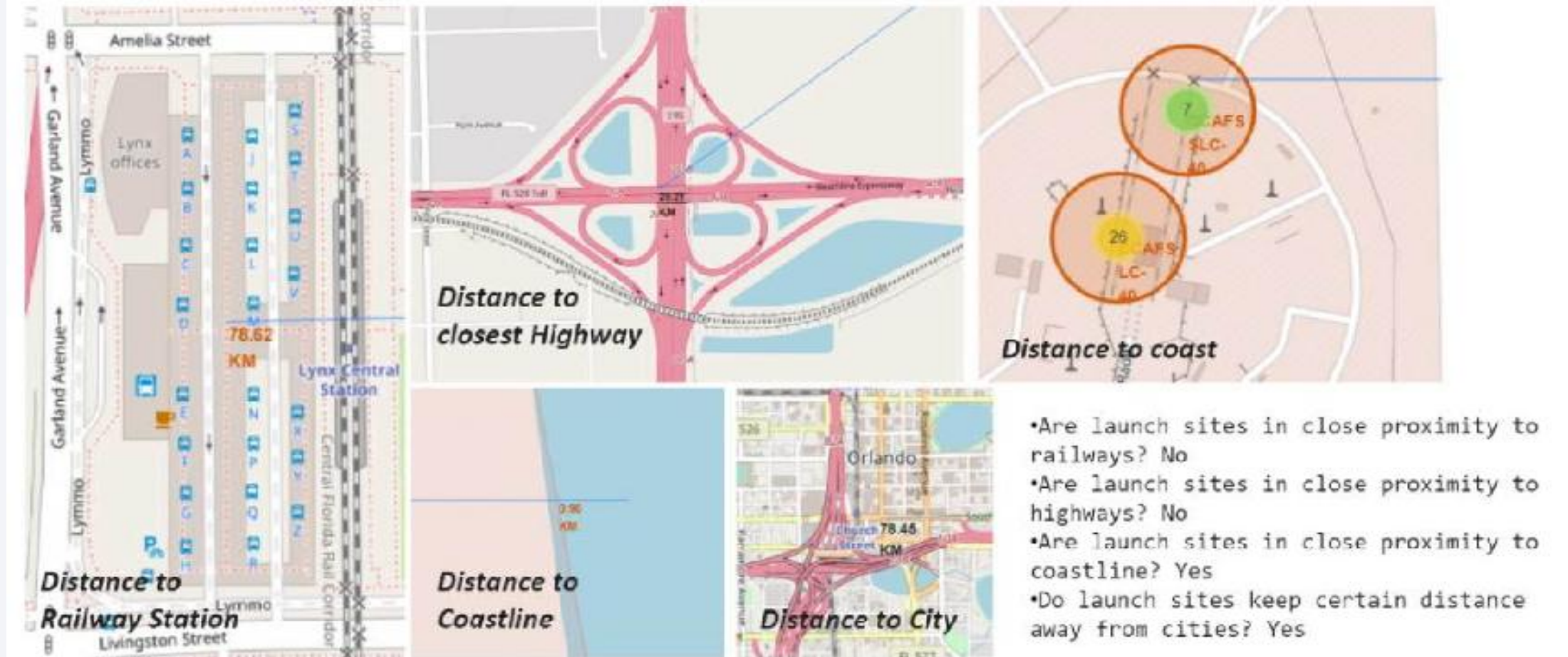
All launch sites global map makers

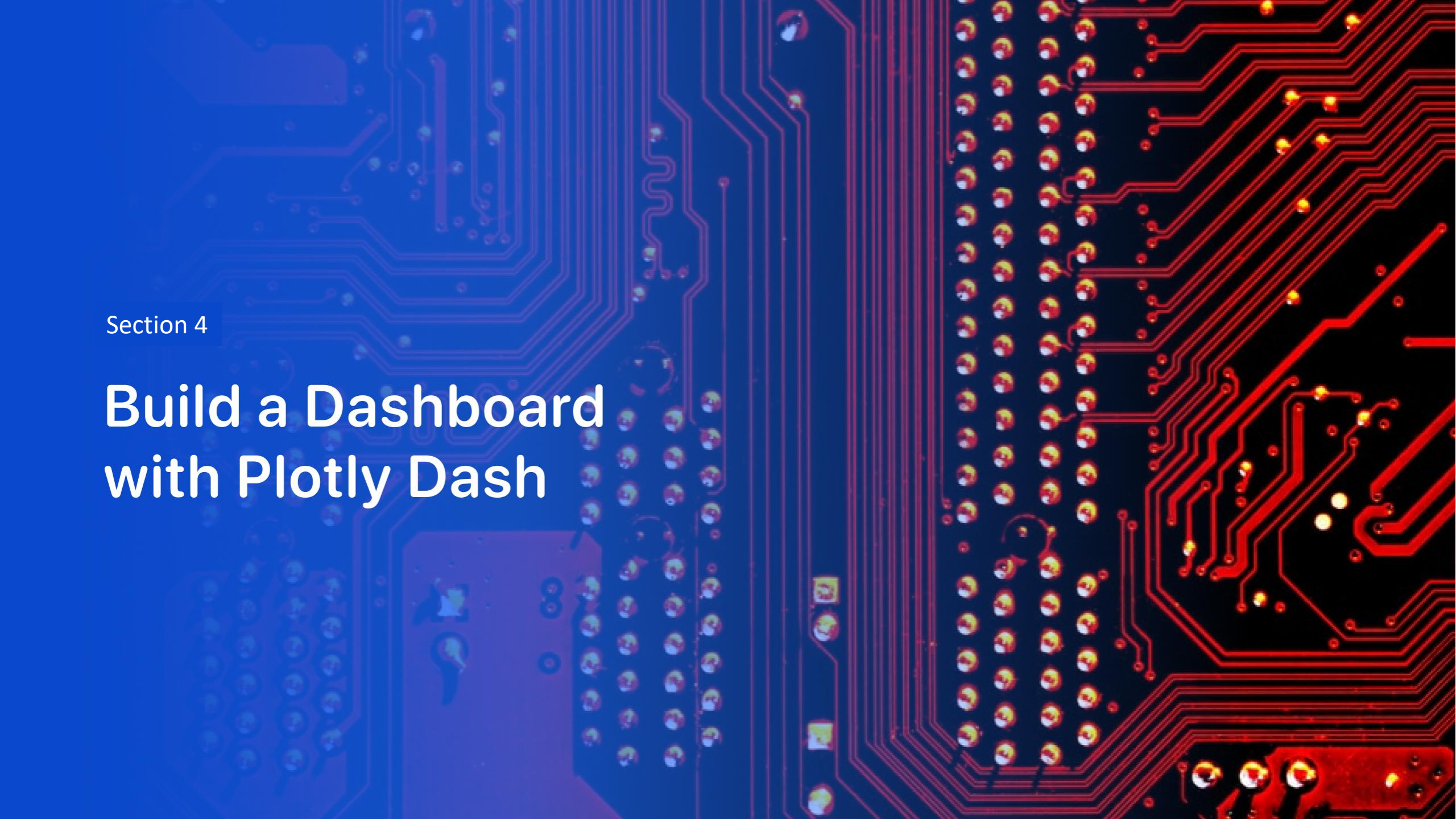


Markers showing launch sites with color labels



Launch sites Distancee to landmarks



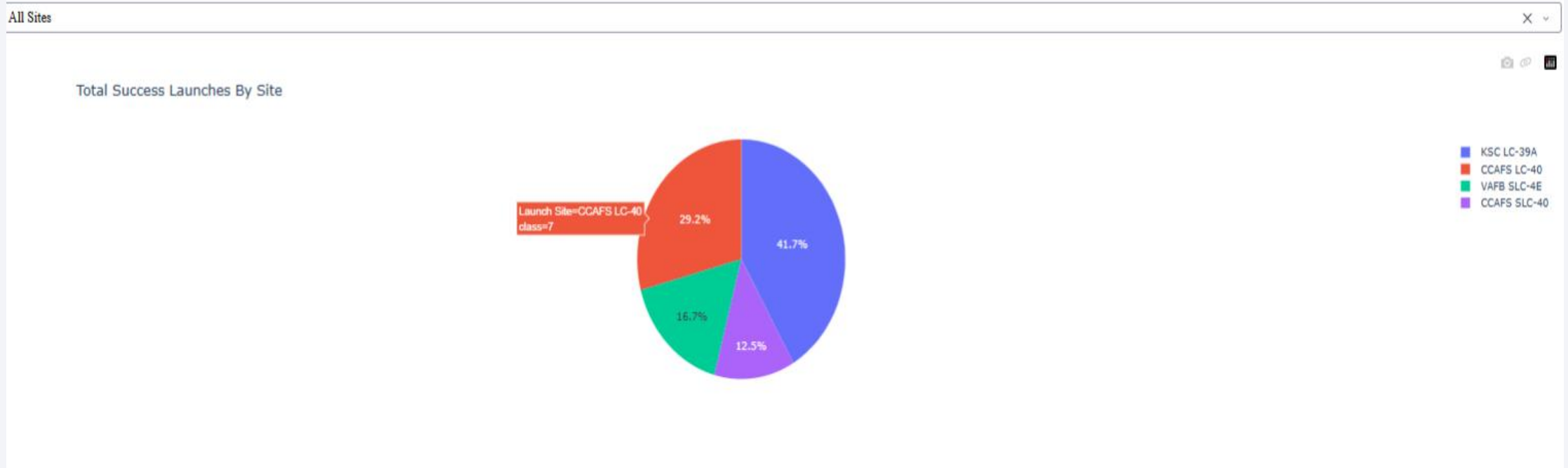


Section 4

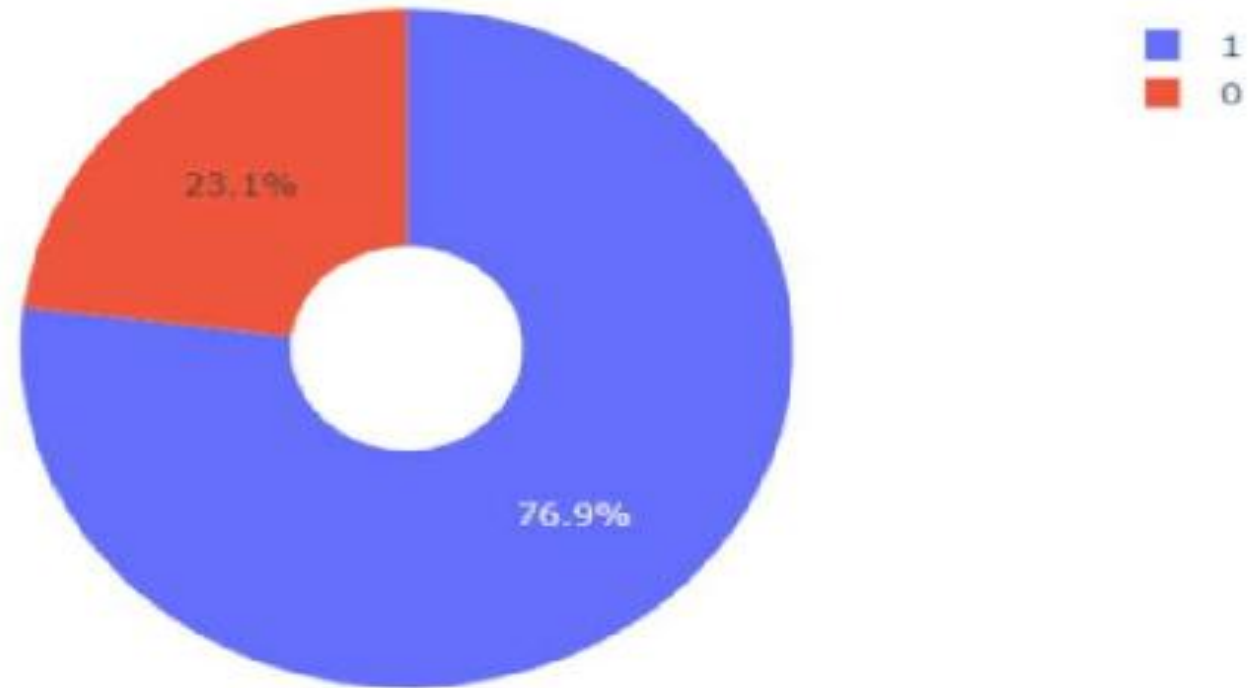
Build a Dashboard with Plotly Dash

Pie chart showing the success percentage achieved by each launch site

We can see that KSC LC-39A has the most successful launches from all the sites

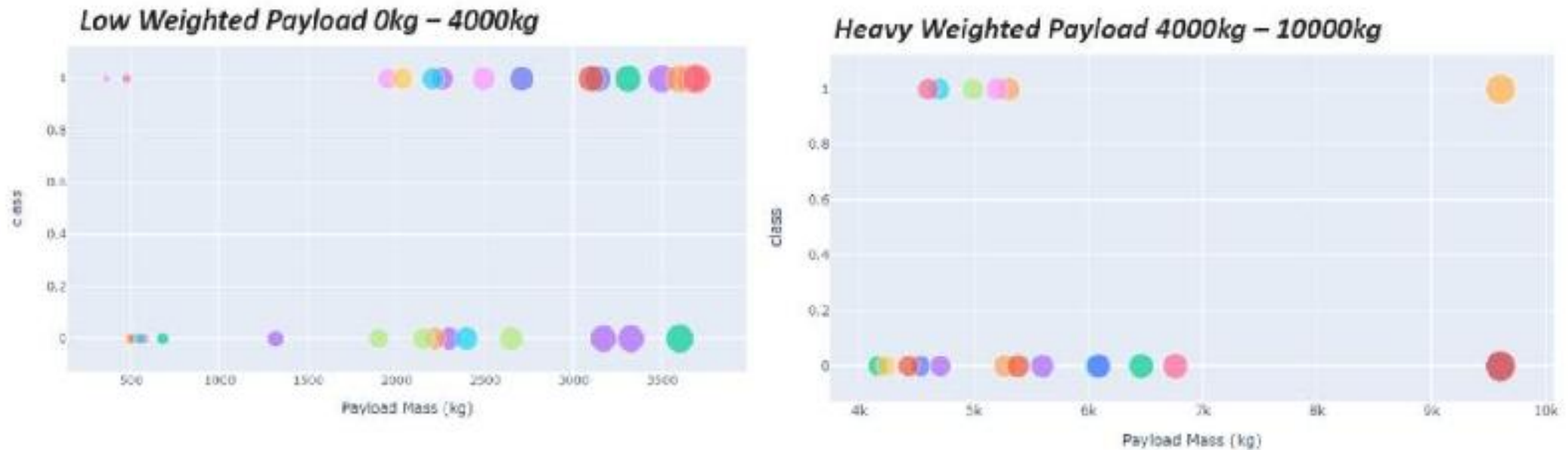


Pie chart showing the launch site with the highest launch success ratio



KSC LC-39A achieved a 76.9% success rate while getting a 23.1% failure rate

Scatter plot of payload vs launch outcome for all sites, with different payload selected in the range slider



We can see the success rates for low weighted payloads is higher than the heavy weighted payloads



Section 5

Predictive Analysis (Classification)

Classification Accuracy

- The decision tree classifier is the model with the highest classification accuracy

In [50]:

```
models = {'KNeighbors':knn_cv.best_score_,
          'DecisionTree':tree_cv.best_score_,
          'LogisticRegression':logreg_cv.best_score_,
          'SupportVector': svm_cv.best_score_}

bestalgorithm = max(models, key=models.get)
print('Best model is', bestalgorithm,'with a score of', models[bestalgorithm])
if bestalgorithm == 'DecisionTree':
    print('Best params is :', tree_cv.best_params_)
if bestalgorithm == 'KNeighbors':
    print('Best params is :', knn_cv.best_params_)
if bestalgorithm == 'LogisticRegression':
    print('Best params is :', logreg_cv.best_params_)
if bestalgorithm == 'SupportVector':
    print('Best params is :', svm_cv.best_params_)
```

```
Best model is DecisionTree with a score of 0.8732142857142856
Best params is : {'criterion': 'gini', 'max_depth': 6, 'max_features': 'sqrt', 'min_samples_leaf':
2, 'min_samples_split': 5, 'splitter': 'random'}
```

Confusion Matrix

- The confusion matrix for the decision tree classifier shows that the classifier can distinguish between the different classes. The major problem is the false postivies .i.e. unsuccessfull landing marked as successful landing by the classifier.



Conclusion

We can conclude that:

- **The larger the flight amount at a launch site, the greater the success rate a launch site.**
- **Launch success rate started to increase in 2013 till 2020.**
- **Orbits ES-L1, GEO, HEO, SSO, VLEO had the most the success rate.**
- **KSC LC-39A had the most successfull launches of any sites.**
- **The Decision tree classifier is the best machine learning algorithm for this task.**

Thank you!

